Table 2 (scenario beta1, beta1_pilot, $B = 48$ per cell). Estimated exposure effects of cumulative total exposure with model misspecification (bold cells with correct model specification).

	Estimated β (true: Cox 1, ERR 1)								Pairs
	True x		Naive		RRC corrected				dropped
$\rho(c, C)$	Cox	ERR	Cox	ERR	Cox.A	Cox.M	ERR.A	ERR.M	(%)
A. Mean $\hat{\beta}$ over converged fits									
Cox+Additive	Cox true $\beta = 1$, additive independent error								
0.3	0.99	1.60	0.62	1.16	**0.95**	1.18	1.77	4.37	
0.6	1.07	2.03	0.70	1.41	**1.05**	1.24	2.23	4.16	
0.9	1.07	2.39	0.71	1.54	**1.07**	1.11	2.31	3.46	
Cox+Multiplicative	Cox true $\beta = 1$, multiplicative independent error								
0.3	1.01	2.98	0.71	1.91	1.13	**1.07**	3.49	3.03	
0.6	0.95	2.29	0.64	1.58	0.94	**0.95**	2.30	2.33	
0.9	1.02	2.50	0.68	1.65	1.02	**1.02**	2.48	2.48	
ERR+Additive	ERR true $\beta = 1$, additive independent error								
0.3	0.67	1.09	0.44	0.72	0.66	0.82	**1.08**	3.48	
0.6	0.80	1.53	0.53	1.00	0.80	0.93	**1.49**	2.35	
0.9	0.74	1.37	0.49	0.93	0.74	0.76	**1.38**	1.59	
ERR+Multiplicative	ERR true $\beta = 1$, multiplicative independent error								
0.3	0.56	1.24	0.45	0.91	0.71	0.68	1.50	**1.39**	
0.6	0.61	1.26	0.43	0.88	0.65	0.64	1.34	**1.31**	
0.9	0.55	1.12	0.37	0.74	0.55	0.55	1.11	**1.12**	
B. Median $\hat{\beta}$ over all fits; [number of fits not converged or stopped at the cap $\beta = 50$]									
Cox+Additive	Cox true $\beta = 1$, additive independent error								
0.3	1.02	1.55 [3]	0.61	1.04 [2]	**0.99**	1.25	1.64 [2]	4.28 [12]	–
0.6	1.09	1.78 [3]	0.73	1.18 [3]	**1.09**	1.32	1.82 [2]	3.38 [4]	–
0.9	1.18	2.11 [1]	0.78	1.42 [2]	**1.17**	1.19	2.16 [2]	2.28	–
Cox+Multiplicative	Cox true $\beta = 1$, multiplicative independent error								
0.3	1.01	2.74	0.71	1.70	1.10	**1.06**	2.66	2.62	–
0.6	0.96	1.95	0.62	1.33	0.95	**0.93**	2.03	1.93	–
0.9	1.05	2.33	0.70	1.50	1.04	**1.03**	2.20	2.25	–
ERR+Additive	ERR true $\beta = 1$, additive independent error								
0.3	0.67	0.93	0.47	0.71 [1]	0.68	0.87	**0.96 [1]**	1.69 [7]	28.5
0.6	0.80	1.09	0.56	0.76	0.83	0.97	**1.21**	1.52 [2]	12.4
0.9	0.70	1.13	0.47	0.72	0.70	0.73	**1.07**	1.11	4.2
ERR+Multiplicative	ERR true $\beta = 1$, multiplicative independent error								
0.3	0.59	1.20	0.48	0.82	0.73	0.72	1.38	**1.25**	0.0
0.6	0.60	1.11	0.42	0.78	0.64	0.64	1.19	**1.17**	0.0
0.9	0.55	0.90	0.36	0.59	0.54	0.54	0.90	**0.90**	0.0

Cox model $\lambda(t) = \lambda_0(t) \exp(\beta x(t))$ with true $\beta_{\text{Cox}} = 1$; ERR model $\lambda(t) = \lambda_0(t)(1 + \beta x(t))$ with true $\beta_{\text{ERR}} = 1$. Each fit is compared with the true β of the model it fits; the paper's values (for $\beta = 1$ and 1.109) are not shown. True x: the same risk model fitted to the true cumulative exposure, separating risk-model misspecification and small-sample bias from measurement error. $n_1 = 2000$, $n_2 = 500$ (at most 8 pairs each), $T^* = 60$, cumulative incidence about 10%. Surrogate C has compound symmetry ($\rho = 0.5$) across years, $E(C) = 0.012$; true $c = \gamma_1 C + \epsilon$ with $\gamma_1 = 2/3$ (additive) or $\log c = \gamma_0 + \log C + \epsilon$ with $E(c) = 0.008$ (multiplicative), σ_ϵ set by $\rho(c, C)$; median x 0.21 (additive) and 0.18 (multiplicative).

Pairs dropped: share of validation pairs with $c \leq 0$ or $C \leq 0$, which the log (.M) calibration cannot use. – : no fit converged, or not recorded. A median of 50.00 means at least half of the fits stopped at the cap $\beta = 50$.